

Configuring every layer 2 switch with “switchport access”. Let’s take as a sample the configuration for ORLANDO.

S1

```
S1# configure terminal
S1(config)#vlan 10
S1(config-vlan)#name management
S1(config-vlan)#exit
S1(config)# interface vlan 10
S1(config-if)#ip address 192.168.1.10 255.255.255.0
S1(config-if)#ip default-gateway 192.168.1.11
S1(config-if)#exit
S1(config)# interface range fa0/1-3
S1(config-if)# switchport access vlan 10
S1(config-if)#exit
S1(config)# interface fa0/4
S1(config-if)#switchport mode trunk
S1(config-if)#end
S1#copy running-config startup-config
```

S2

```
S2# configure terminal
S2(config)#vlan 20
S2(config-vlan)#name internet
S2(config-vlan)#exit
S2(config)# interface vlan 20
S2(config-if)#ip address 192.168.99.20 255.255.255.0
S2(config-if)#ip default-gateway 192.168.99.21
S2(config-if)#exit
```

```
S2(config)# interface range fa0/1-2
S2(config-if)# switchport access vlan 20
S2(config-if)#exit
S2(config)# interface fa0/3
S2(config-if)#switchport mode trunk
S2(config-if)#end
S2#copy running-config startup-config
```

S3

```
S3# configure terminal
S3(config)#vlan 30
S3(config-vlan)#name production
S3(config-vlan)#exit
S3(config)# interface vlan 30
S3(config-if)#ip address 192.168.11.30 255.255.255.128
S3(config-if)#ip default-gateway 192.168.11.31
S3(config-if)#exit
S3(config)# interface range fa0/2-4
S3(config-if)# switchport access vlan 30
S3(config-if)#exit
S3(config)# interface fa0/1
S3(config-if)#switchport mode trunk
S3(config-if)#end
S3#copy running-config startup-config
```

Configuring every layer 3 switch with “trunk mode”. Commands used in configuring “Multilayer Switch (L3) on ORLANDO site”:

L3OrlandoSwitch

```
OrlandoSwitch# configure terminal
OrlandoSwitch(config)#vlan 10
OrlandoSwitch(config-vlan)#name management
OrlandoSwitch(config-vlan)#exit
OrlandoSwitch(config)# interface fa0/1
OrlandoSwitch(config-if)#switchport mode trunk
OrlandoSwitch(config-if)#exit
OrlandoSwitch(config)#interface vlan 10
OrlandoSwitch(config-if)#ip address 192.168.1.11 255.255.255.0
OrlandoSwitch(config-if)#ip default-gateway 192.168.1.11
OrlandoSwitch(config-if)#end
OrlandoSwitch# copy running-config startup-config
```

```
OrlandoSwitch# configure terminal
OrlandoSwitch(config)#vlan 20
OrlandoSwitch(config-vlan)#name internet
OrlandoSwitch(config-vlan)#exit
OrlandoSwitch(config)# interface fa0/2
OrlandoSwitch(config-if)#switchport mode trunk
OrlandoSwitch(config-if)#exit
OrlandoSwitch(config)#interface vlan 20
OrlandoSwitch(config-if)#ip address 192.168.99.21 255.255.255.0
OrlandoSwitch(config-if)#ip default-gateway 192.168.99.21
OrlandoSwitch(config-if)#end
OrlandoSwitch# copy running-config startup-config

OrlandoSwitch# configure terminal
OrlandoSwitch(config)# interface vlan 30
OrlandoSwitch(config-vlan)#name production
```

OrlandoSwitch(config-vlan)#exit
OrlandoSwitch(config)# interface fa0/3
OrlandoSwitch(config-if)#switchport mode trunk
OrlandoSwitch(config-if)#exit
OrlandoSwitch(config)#interface vlan 30
OrlandoSwitch(config-if)#ip address 192.168.11.31 255.255.255.128
OrlandoSwitch(config-if)#ip default-gateway 192.168.11.31
OrlandoSwitch(config-if)#end
OrlandoSwitch# copy running-config startup-config
OrlandoSwitch#interface fa0/4
OrlandoSwitch(config-if)#switchport trunk encapsulation dot1q
OrlandoSwitch(config-if)#switchport mode trunk

**Configuring every layer 3 switch and every router on the site with “router on a stick”.
Commands used in configuring “Router1 on ORLANDO site”:**

R1# configure terminal
R1(config)# interface fa0/0
R1(config-if)#ip address 192.168.5.2 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)# interface fa0/0.10
R1(config-if)#encapsulation dot1q 10
R1(config-if)#ip address 192.168.1.11 255.255.255.0
R1(config-if)# interface fa0/0.20
R1(config-if)#encapsulation dot1q 20
R1(config-if)#ip address 192.168.99.21 255.255.255.0
R1(config-if)# interface fa0/0.30
R1(config-if)#encapsulation dot1q 30

R1(config-if)#ip address 192.168.11.31 255.255.255.128

R1(config-if)#end

R1# copy running-config startup-config